Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 19)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 1 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 1 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 2 Table 1 compares organic and inorganic fertilisers.

0 2 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
~~Inorganic~~ (eg ammonium nitrate, NPK)		

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0 2 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 3 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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2 .....

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0 3 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 3 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 4 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 4 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 4 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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0 5 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 5 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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0 5 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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0 5 . 3 State the equation used to forecast a change in population size.

[1 mark]

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0 5 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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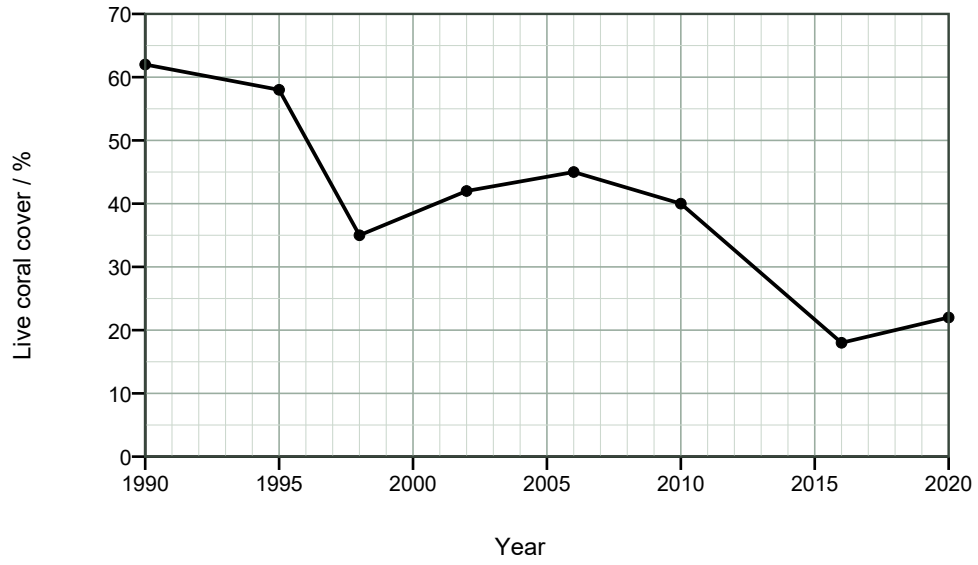
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0 6 Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 6 . 1 Describe the changes in coral cover shown in Figure 1.

[2 marks]

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0 6 . 2 Explain what coral bleaching is and give **three** causes.

[4 marks]

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0 6 . 3 Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

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0 6 . 4 Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

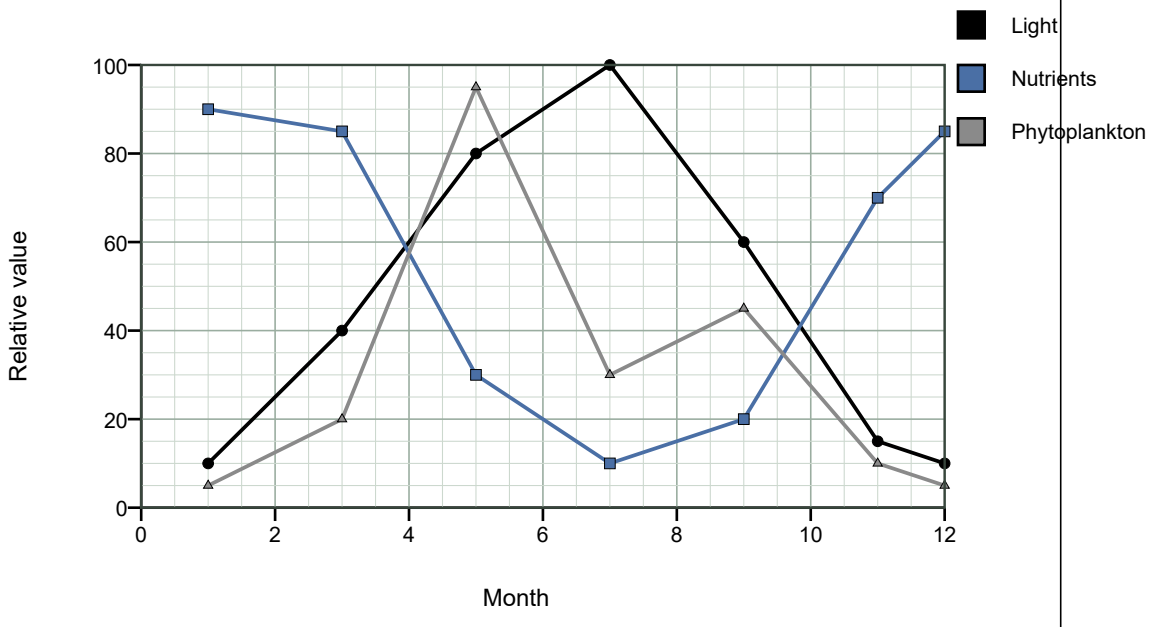
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0 7 Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



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0 7 . 1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

0 7 . 2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

0 7 . 3 Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 7 . 4 Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

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0 8 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

0 8 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

0 8 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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0 9 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 9 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

Answer

0 9 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

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0 9 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

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0 9 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

0 9 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

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1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS